

Data Flow Diagram

①.

- A graphical tool, useful for communicating with users, managers & other personnel.
- Useful for ~~existing~~ analyzing existing as well as proposed systems
- Mainly focus on the movement of data b/w external entities & processes, & between processes & data stores.
- Simple technique.

Why DFD is important ?

②.

- What data are stored
- What data a system processes.
- What Transformations are performed.
- What results are produced.
- Graphical nature make it a good communication tool b/w [user & Analyst.]
& [Analyst & System Designer]

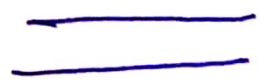
DFD Elements

③

- Source / Sinks (External Entities) → e.g. any used.
(use something, like customer).
Represented by 

- Data Flow
In what direction your data flow. (move).

Process :- represented by  Circle
What work performed.

Data store database used  (data store)
(read DB, Write DB)

External Entities

(4)

- A Rectangle represents an external entity.
- They either supply or receive data.
- They do not process data.

(1/p). Source — Entity that supplies data to the system.

Sink — Entity that receives data from the system.

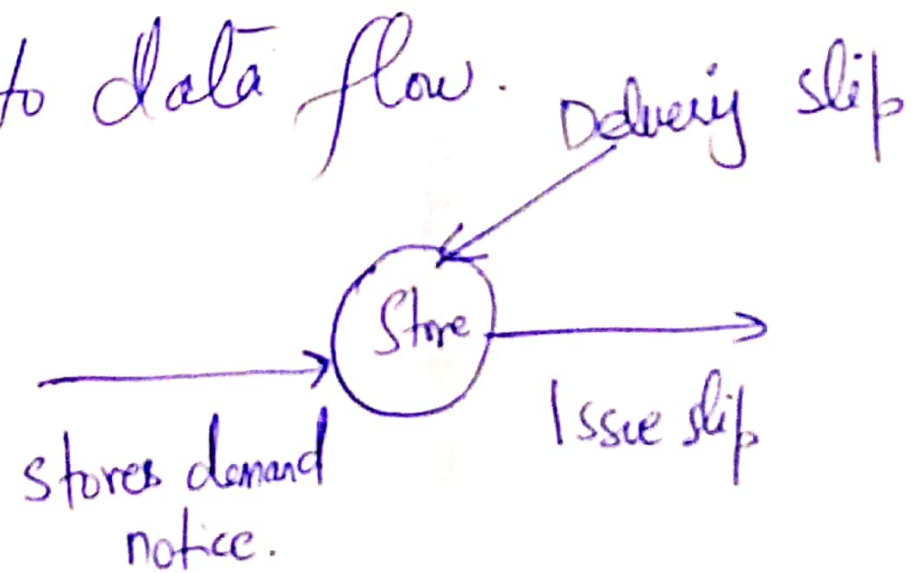
Data Flow

- Movement of data through the system
- Connects the processes, external entities & data stores -
- Generally unidirectional, If same data flows in both directions, double headed arrows can be used.

Process

(6)

- Circle represents a process.
- Straight line with incoming arrows are input data flows.
- Straight lines with outgoing arrows are o/p data flows.
- Labels are assigned to data flow.

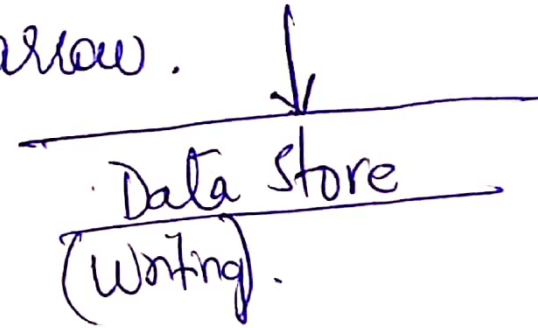


Data stores

Data store is a repository of data.

Data store

Data can be written into the Data store. This is depicted by an incoming arrow.



Data can be read from a data store. Depicted by an outgoing arrow.



External entity cannot read or write to the data store.

Data flow Rules

(8)

Data can flow from.

- External entity to process.
- Process to external entity.
- Process to store & back.
- Process to process.

Data cannot flow from

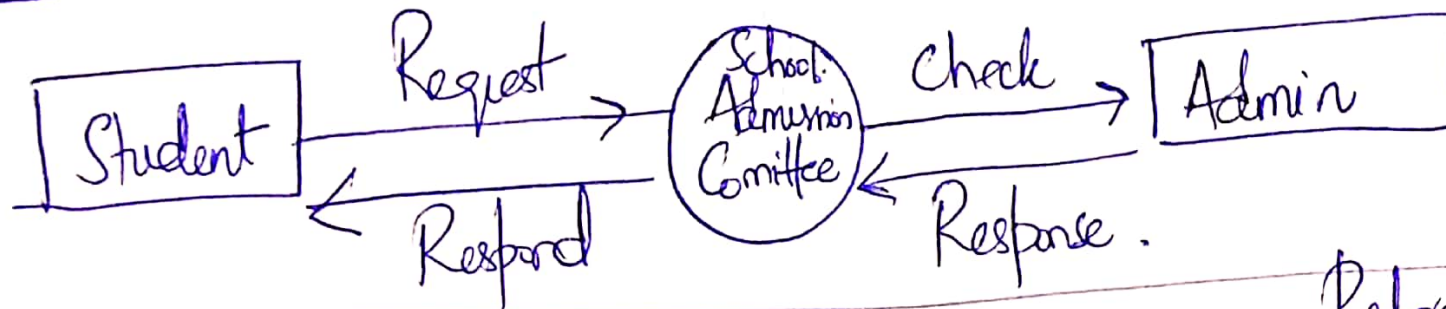
- External entity to external entity.
- External entity to store.
- Store to external entity.
- Store to store.

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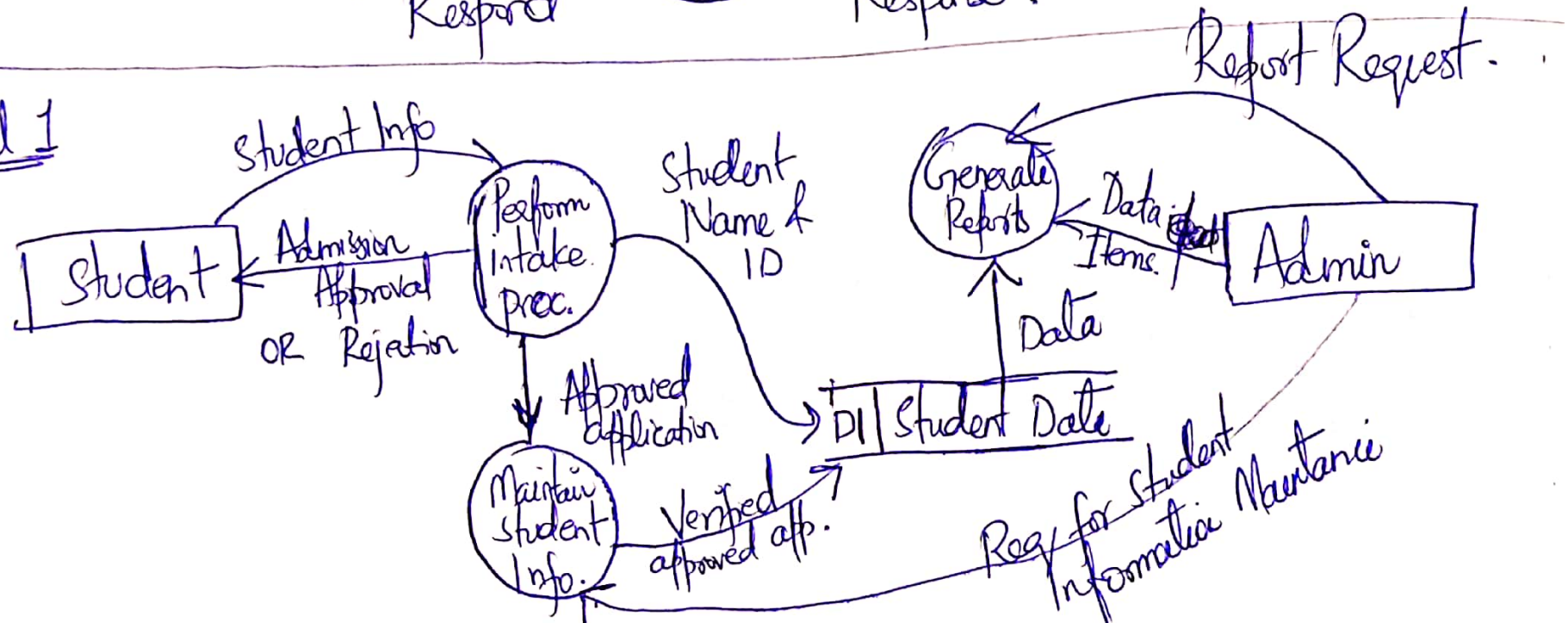
Level of DFD

Three level of DFD, level 0, level 1, level 2

Level 0

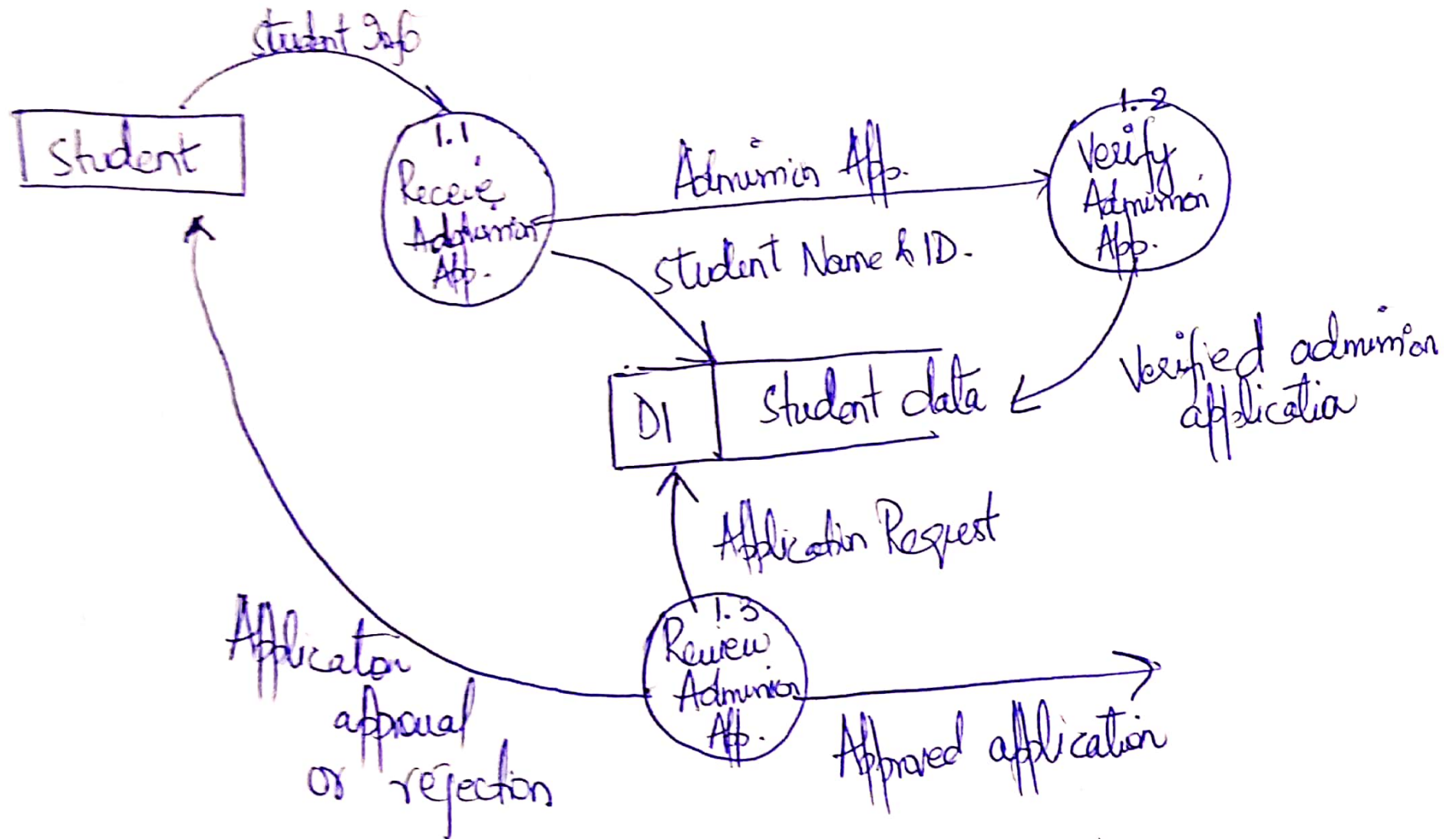


Level 1



Level 2 Process 1 Perform Intake Procedure

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logical DFD $\rightarrow$ ^{what we} trying to do, Generalize, not about Implementation (11)
Physical DFD $\rightarrow$ Implementation, Uses of DFD.

- DFD considered as so far are called logical DFD's.
- A physical DFD is similar to a document flow diagram.
- It specifies who does the operations specified by the logical DFD.
- Physical DFD may depict physical movements of goods.
- Physical DFD can be drawn during fact gathering phase of a life cycle.

Physical DFD for Cheque Encashment

(12).

